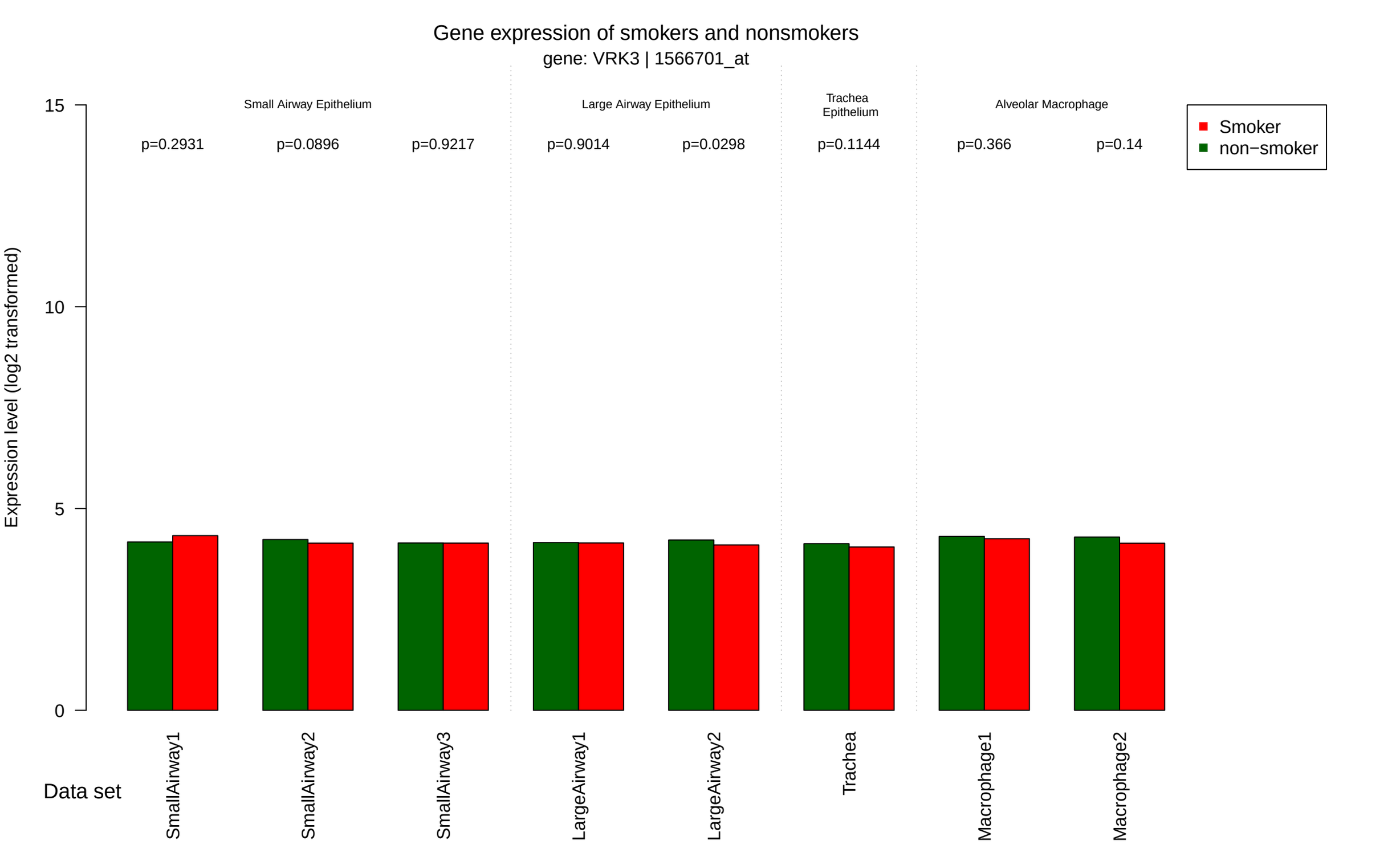
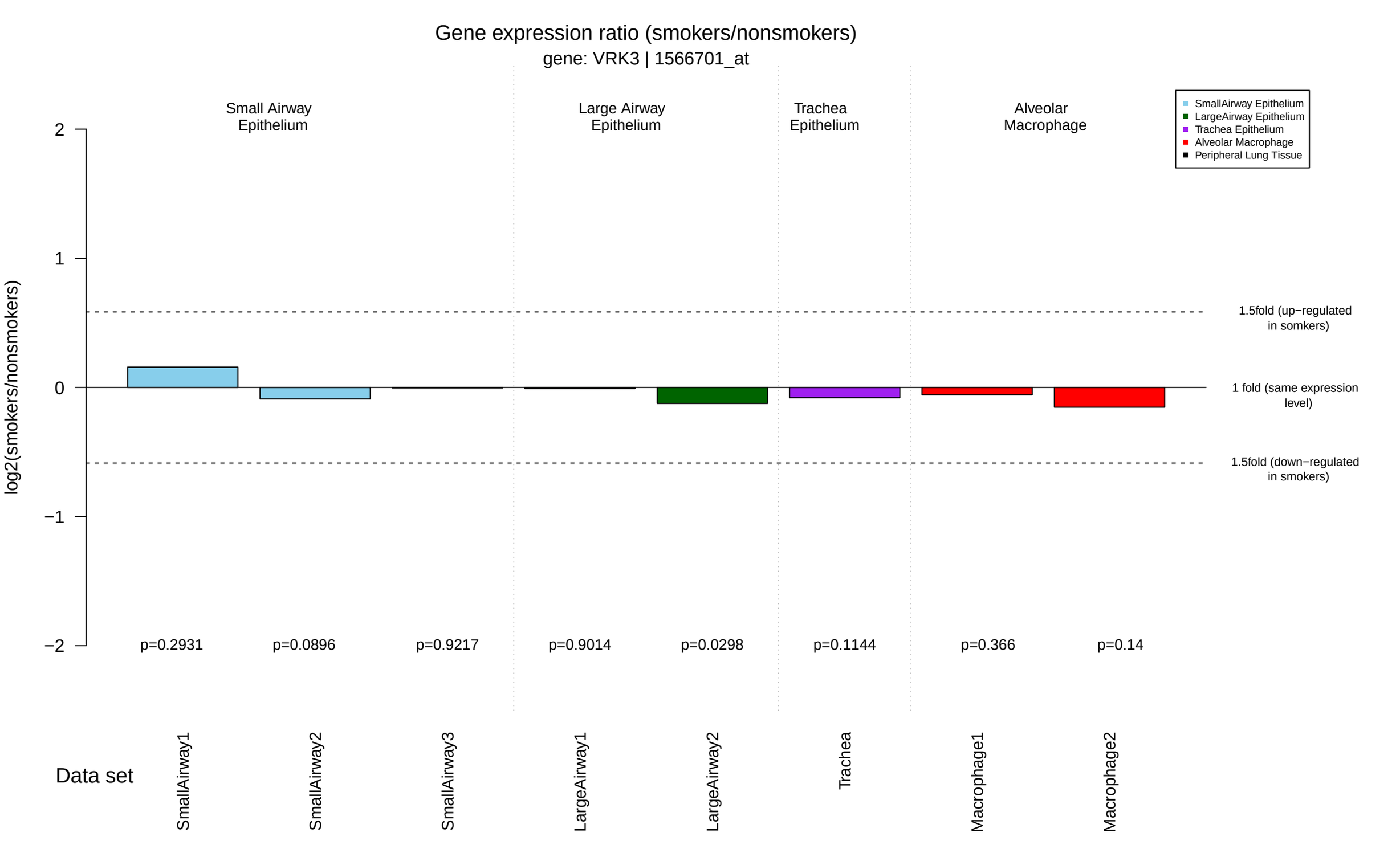
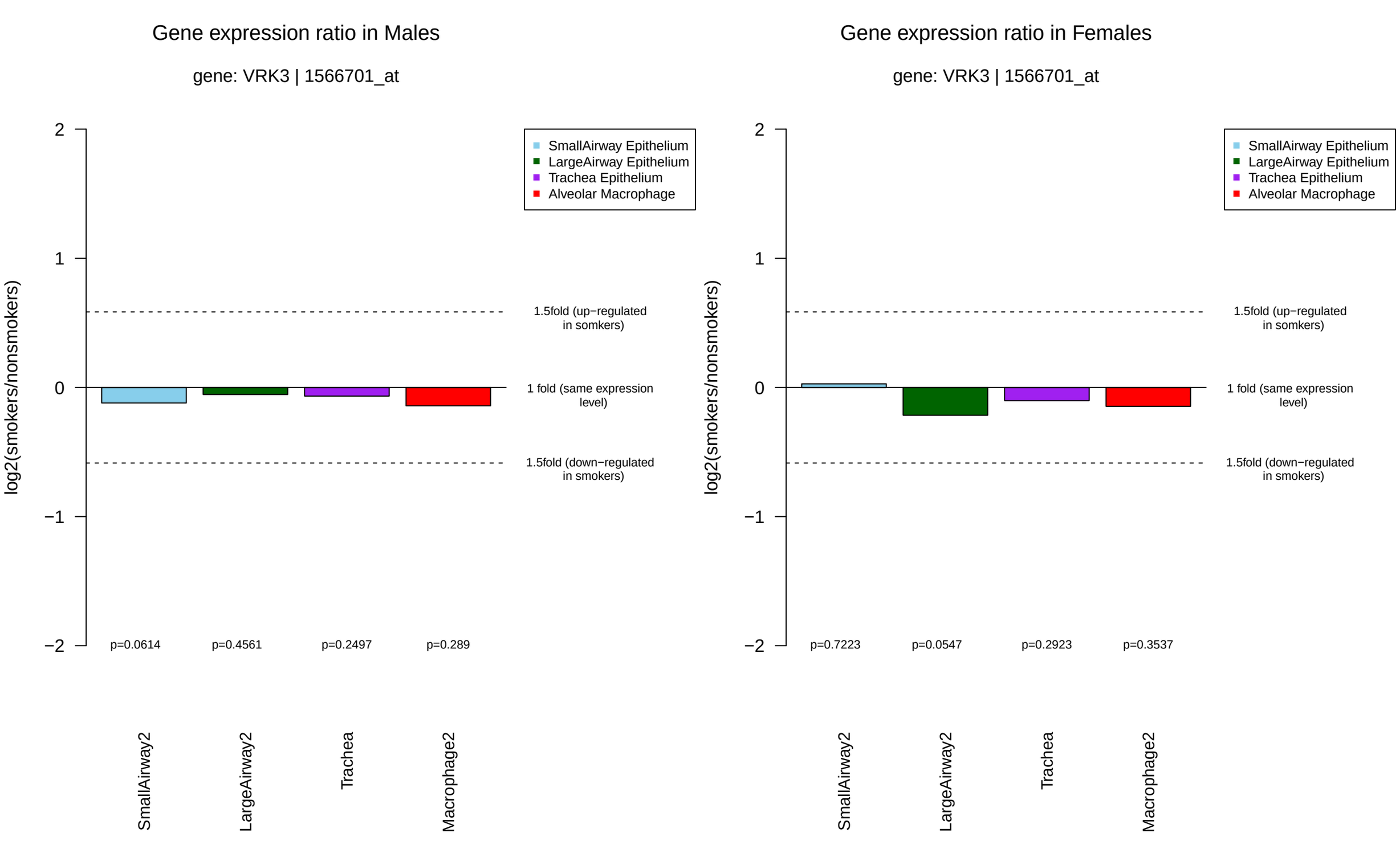
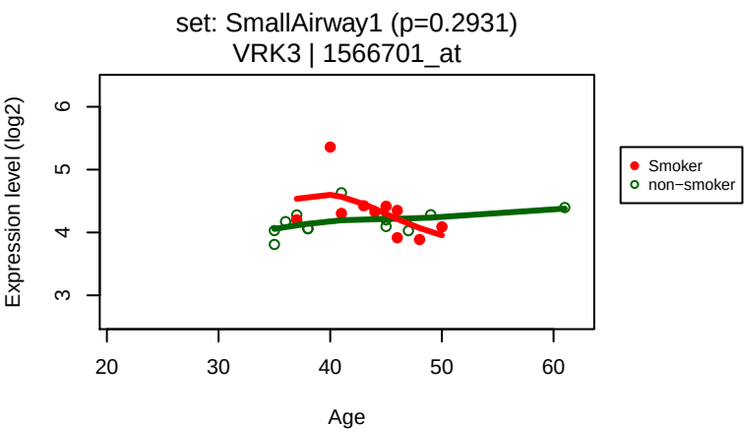
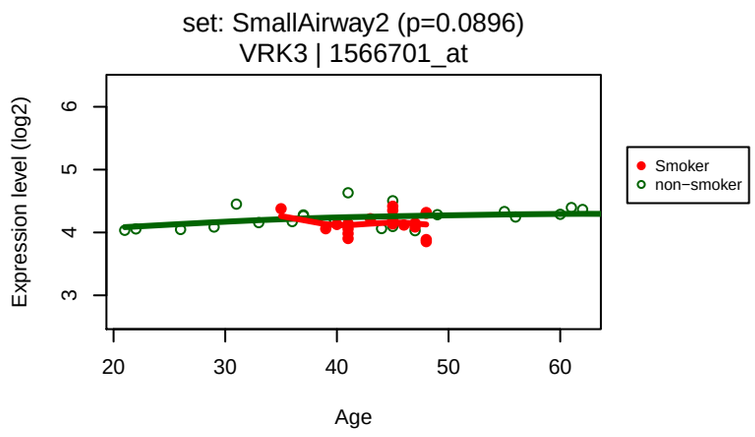
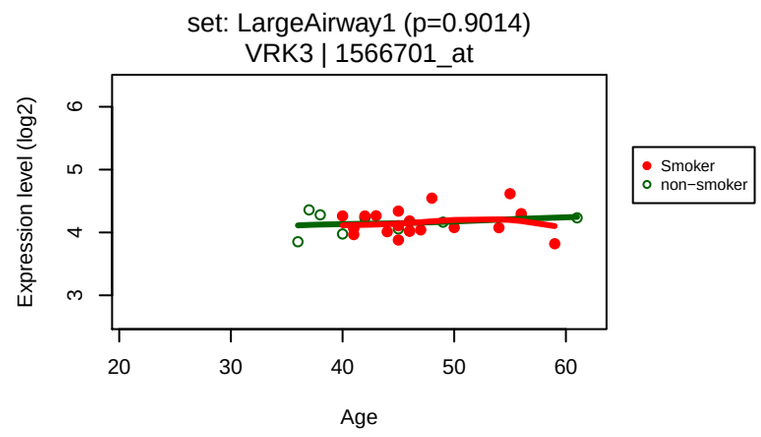

Gene expression of smokers and nonsmokers
gene: VRK3 | 1566701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2931
p=0.0896
p=0.9217
p=0.9014
p=0.0298
p=0.1144
p=0.366
p=0.14
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

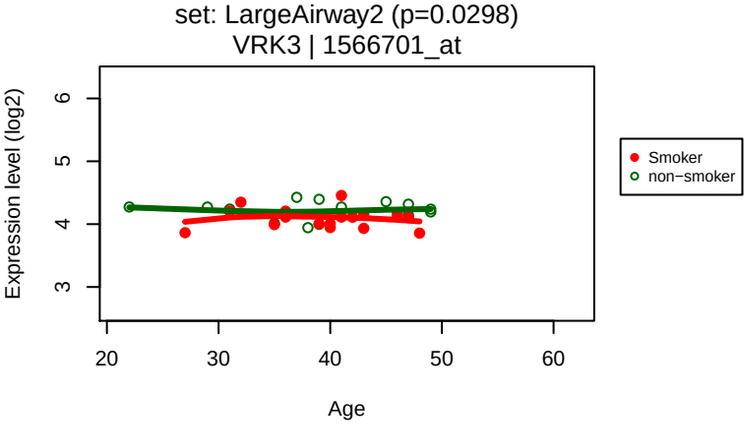

Gene expression ratio (smokers/nonsmokers)
gene: VRK3 | 1566701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2931
p=0.0896
p=0.9217
p=0.9014
p=0.0298
p=0.1144
p=0.366
p=0.14
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

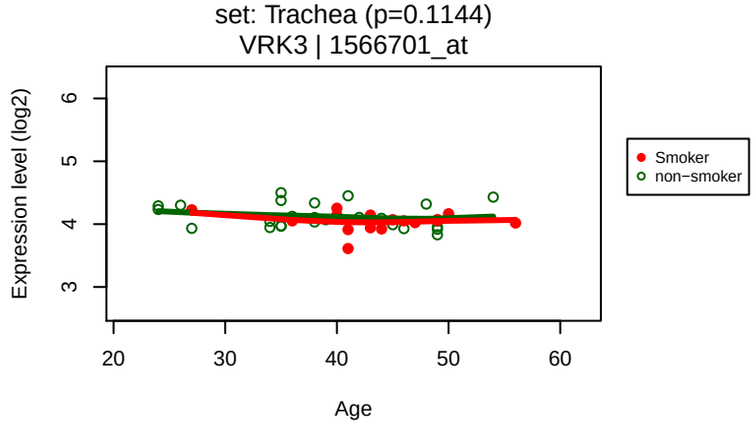

Gene expression ratio in Males
gene: VRK3 | 1566701_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0614
p=0.4561
p=0.2497
p=0.289
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VRK3 | 1566701_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7223
p=0.0547
p=0.2923
p=0.3537
SmallAirway2
LargeAirway2
Trachea
Macrophage2

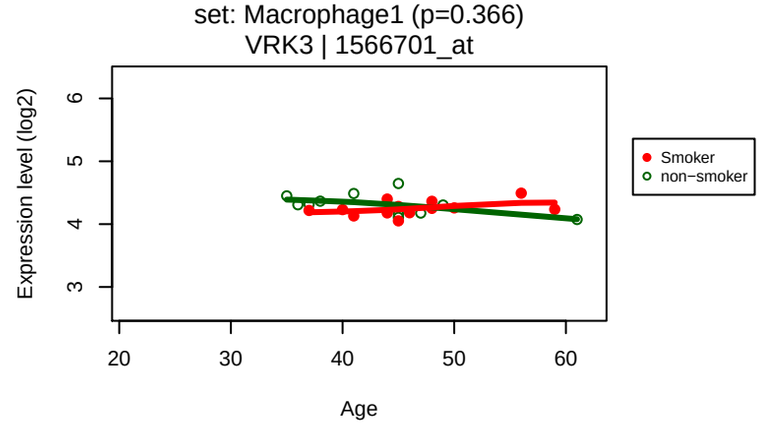

set: SmallAirway1 (p=0.2931)
VRK3 | 1566701_at
Expression level (log2)
Age
Smoker
non−smoker

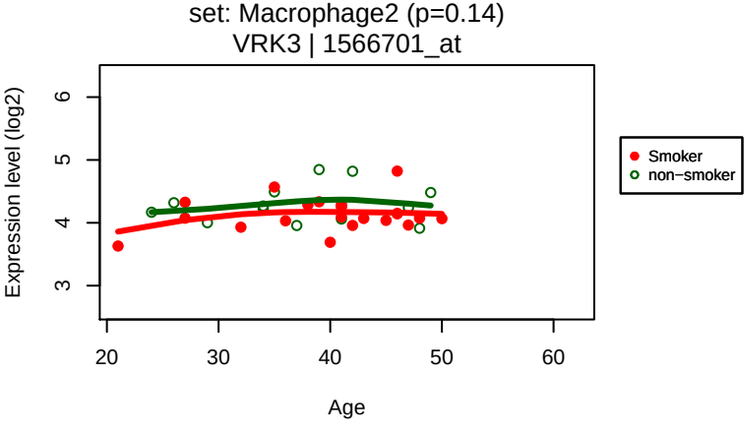

set: SmallAirway2 (p=0.0896)
VRK3 | 1566701_at
Expression level (log2)
Age
Smoker
non−smoker

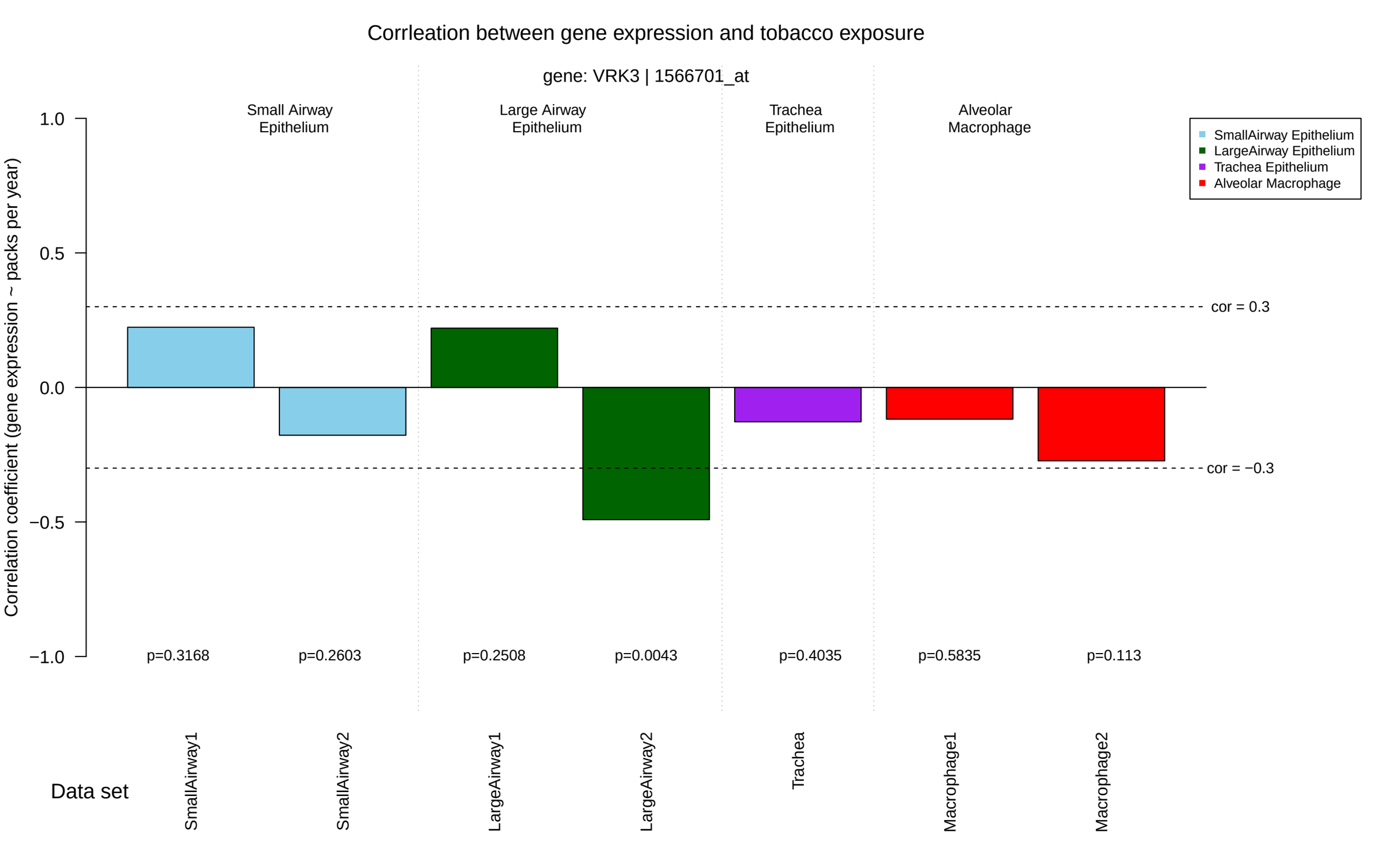

set: LargeAirway1 (p=0.9014)
VRK3 | 1566701_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0298)
VRK3 | 1566701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1144)
VRK3 | 1566701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.366)
VRK3 | 1566701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.14)
VRK3 | 1566701_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VRK3 | 1566701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3168
p=0.2603
p=0.2508
p=0.0043
p=0.4035
p=0.5835
p=0.113
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2